LeChristopher Blackwell

Senior Software Engineer, Applied AI

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Summary

Senior software engineer, 8 years in production, nearly seven at AvantLink on a multi-developer data platform, building AI products in Python, TypeScript/React, and AWS. Takes underspecified problems to live systems and measures the result: every model output a reader sees is gated by deterministic checks, typed citations, or an explicit refusal. Eleven live systems, one at 7.4M requests a week, tools on PyPI and Homebrew, fixes merged in SQLFluff and SpikeInterface.

Skills

Languages & Backend: Python, TypeScript, JavaScript, SQL; FastAPI, Node.js/Express, REST, SSE, and WebSocket APIs
AI Systems: agent orchestration, MCP servers, RAG, evals and verification, cost-tiered model routing, Amazon Bedrock, Claude Code
Frontend & Data: React 18/19, React Router, TanStack, Vite, D3; PostgreSQL, Neo4j, Redis, Snowflake, Redshift, SQLite
Infrastructure: AWS (ECS/Fargate, RDS, S3, CloudFront, VPC), Docker, Kubernetes, OpenTofu/Terraform, GitHub Actions

Professional Experience

Fourth Axis Jun 2026 - Present
Technical Consultant • three defense acquisition systems • TypeScript, Python/FastAPI, React, Anthropic API, AWS

- Built an AI drafting workbench for defense acquisition documents with the domain expert in the loop by design: model suggestions land pending in a review UI, provenance separates retrieved facts from model recall, and a document completes only after a person has accepted every pending field
- Measured every prompt and model change with an interleaved A/B eval harness after catching session-to-session drift that moved a byte-identical prompt's citation rate from 30% to 87%
- Built an evidence graph ranking thousands of US research institutions against a client's priority mission areas: a fully deterministic ranking over public data, plus independent guards and typed citations on every generated claim
- Shipped a precedent-analysis tool for federal solicitations, every result resolving to its source paragraph in a public-record corpus: React 19 over FastAPI with vector retrieval, deployed to AWS via GitHub Actions

MethodProof, Inc. Jul 2025 - Present
Founder & Lead Engineer • FastAPI, React, Neo4j, PostgreSQL, Redis, AWS • methodproof.com, closed alpha

- Founded and built an engineering process intelligence platform that records how engineers work with AI tools and models it as a Neo4j knowledge graph: FastAPI API, React dashboard, browser extension, and mobile client, each containerized with its own CI/CD pipeline; three provisional patents filed
- Engineered the ingestion and model layers: Redis Streams processing 42 event types into Neo4j behind a client-side buffer that survives connectivity drops, and one MCP server wrapping all LLM access with model allowlists, per-session token budgets, and a full prompt and completion log
- Deployed the full AWS stack through OpenTofu: a three-tier VPC with ECS Fargate, RDS PostgreSQL, ElastiCache Redis, Neo4j on EC2, S3, and CloudFront

AvantLink Oct 2020 - Jun 2025
Fullstack Software Engineer • Python ETL, Redshift, Dremio, AWS Batch/Lambda • Park City, UT

- Designed and shipped an LLM-powered reporting application from concept to internal deployment for business stakeholders: prompt engineering, structured output parsing, and integration with the platform's existing data APIs
- Built and maintained internal API services and data pipelines, cutting data discrepancy detection time by 75% through automated validation across Redshift, Snowflake, and third-party APIs
- Owned API integration contracts with international partner engineering teams across time zones: technical specification, implementation, schema normalization, retries, and alerting on pipeline failures
- Shipped order-pipeline optimizations, custom reporting, and currency handling into an established, multi-developer data warehouse platform through the team's ticket and code review process

AvantLink Oct 2018 - Oct 2020
Software Engineer, Testing Focus • Python, Snowflake • Park City, UT

- Built Python automation services for data warehouse validation, cutting manual verification cycles by 30% and improving data accuracy across Snowflake environments
- Developed API integration testing that validated data flow across distributed systems, accelerating release cycles by 25%

Earlier Nov 2016 - Oct 2018

Automation Engineer, Openside (Provo, UT) • *Frontend Software Engineer, Stylisted (Chicago, IL)*

- Built workflow automation systems and database architectures supporting business process pipelines; led a CRM integration, shipping React/Redux features and reducing technical debt

Projects

Academic Ancestry 2026 - Present

Python, PostgreSQL, React 19 SSR • *MCP server with OAuth* • *AWS (EC2, CloudFront, S3, Bedrock)* • *academicancestry.org*

- Built and shipped alone an academic genealogy over open data, public within ten days of its first measured layer: 6.0M theses harvested nightly from 449 repositories in 57 countries, 14 metadata formats, plus Wikidata's advisor graph
- Designed the evidence model so no model draws a line: every link is cited from a record, extracted from a thesis PDF by a model and promoted only through a measured gate, or inferred and drawn but never counted; two spellings stay two people until a measured match rule says otherwise, and a branch that stops because the data stops says so
- Took one small server from 1.1M to 7.4M requests a week (2.25M in a day) by metering what each request costs, with per-day spend caps on the paid API by caller class; shipped an OAuth-gated MCP server so Claude or ChatGPT query the record with evidence attached, where "not covered" and "no relation" are distinct answers

mAiSight 2026 - Present

React 18, TanStack Router + Query, Vite • *Python, FastAPI, DuckDB, Anthropic* • *maisight.codeblackwell.ai*

- Built bidirectional glossary highlighting for jargon-heavy documents: Markdown, MDX, OpenAPI, or LaTeX in, a reading page out where every defined term lights up in the prose and in a scroll-synced glossary rail; an embed widget and browser extension carry the same mechanic into existing docs sites
- Made model choice measurable: term extraction runs against Anthropic or any OpenAI-compatible endpoint and must pass a gold-fixture eval before a model is trusted, and every extracted term passes human accept, edit, or reject review before it is published
- Built reader analytics from observed counts, deleting a composite confusion score after measuring its 0.88 correlation with raw traffic; ten subdomains served from one container, with CI shipping every green push

crg-debug 2026 - Present

Public Claude Code plugin, MIT • *github.com/CodeBlackwell/crg-debug*

- Packaged eight graph-driven engineering workflows around one idea: build a knowledge graph of the codebase, fan single-purpose agents out over non-overlapping slices of it, and let deterministic code decide when the work is done
- Engineered the trust model: independent refuter and reproducer agents verify every finding, gate agents judge on raw exit codes alone, and human approval sits at every consequential boundary; proved against production code, where it surfaced and root-caused the four upstream bugs under Open Source below

Open Source & Agentic Tooling

- **PANEL**: deployed 18-agent PRD debate system (AutoGen GroupChat, SSE streaming); dual-model cost tiering cut per-session cost 65% ($0.34 to $0.12) | panel.codeblackwell.ai
- **PROVE**: ReAct agent over a code knowledge graph; dual-provider embeddings with automatic fallback, and cost-tiered routing A/B tested to quality parity at 2.9x lower cost and 2.2x faster | prove.codeblackwell.ai
- **Upstream contributions**: root-caused fixes with tests in four unfamiliar codebases; merged into SQLFluff and SpikeInterface (Python), in review at OpenTelemetry collector-contrib (Go), superseded by the maintainer's own fix at Moq.Analyzers (C#/Roslyn)
- **Agentopolis**: real-time Claude Code agent-fleet visualizer on PyPI + Homebrew (MIT); repos render as an isometric city while a dispatch floor animates live agent activity via hooks | agentopolis.codeblackwell.ai
- **Vizardry**: Claude Code plugin (MIT) for React + D3 charts: 212 exemplars, five verifiers rendering each chart in jsdom and failing prose with a number the data never computed, and a gated parallel build where no agent certifies its own card; retrieval plus verification beat a fine-tune, rubric 50% to 80% | github.com/CodeBlackwell/vizardry

Education

IBM Generative AI Engineering Specialist Certification (2025) • **Hack Reactor**, Software Engineering (2016)